

CSCI 210: Computer Architecture

Lecture 14: Digital Logic

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Slides from Cynthia Taylor

CS History: The Manchester Transistor Computer

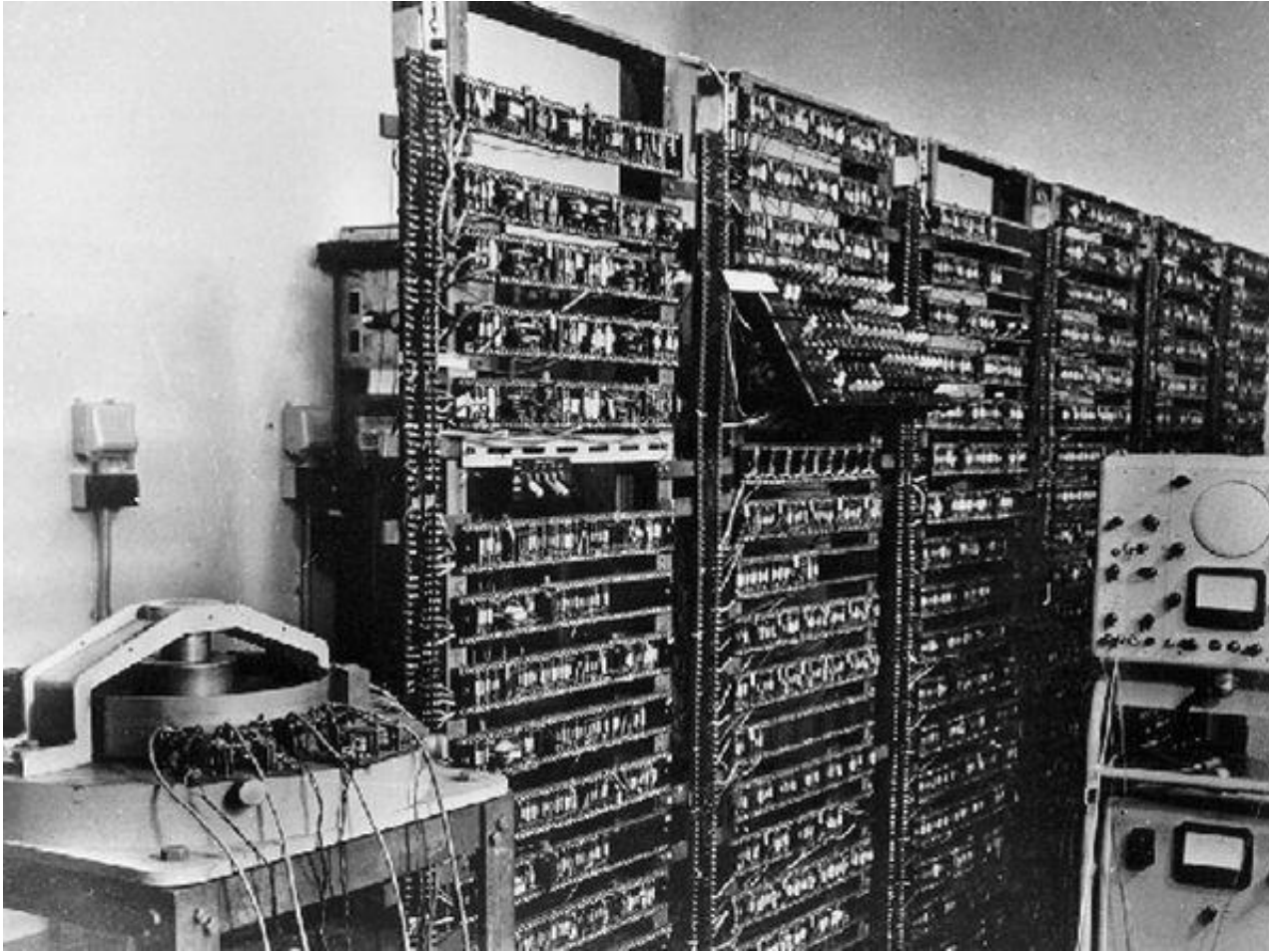


Image credit: The University of Manchester

- First computer to use transistors
- Developed at University of Manchester in 1953
- Problems with the reliability of early batches of transistors meant that its mean time between failures was about 90 minutes
- Still used vacuum tubes for its clock and memory, so not fully transistorized

Creating the Universe from 1 and 0

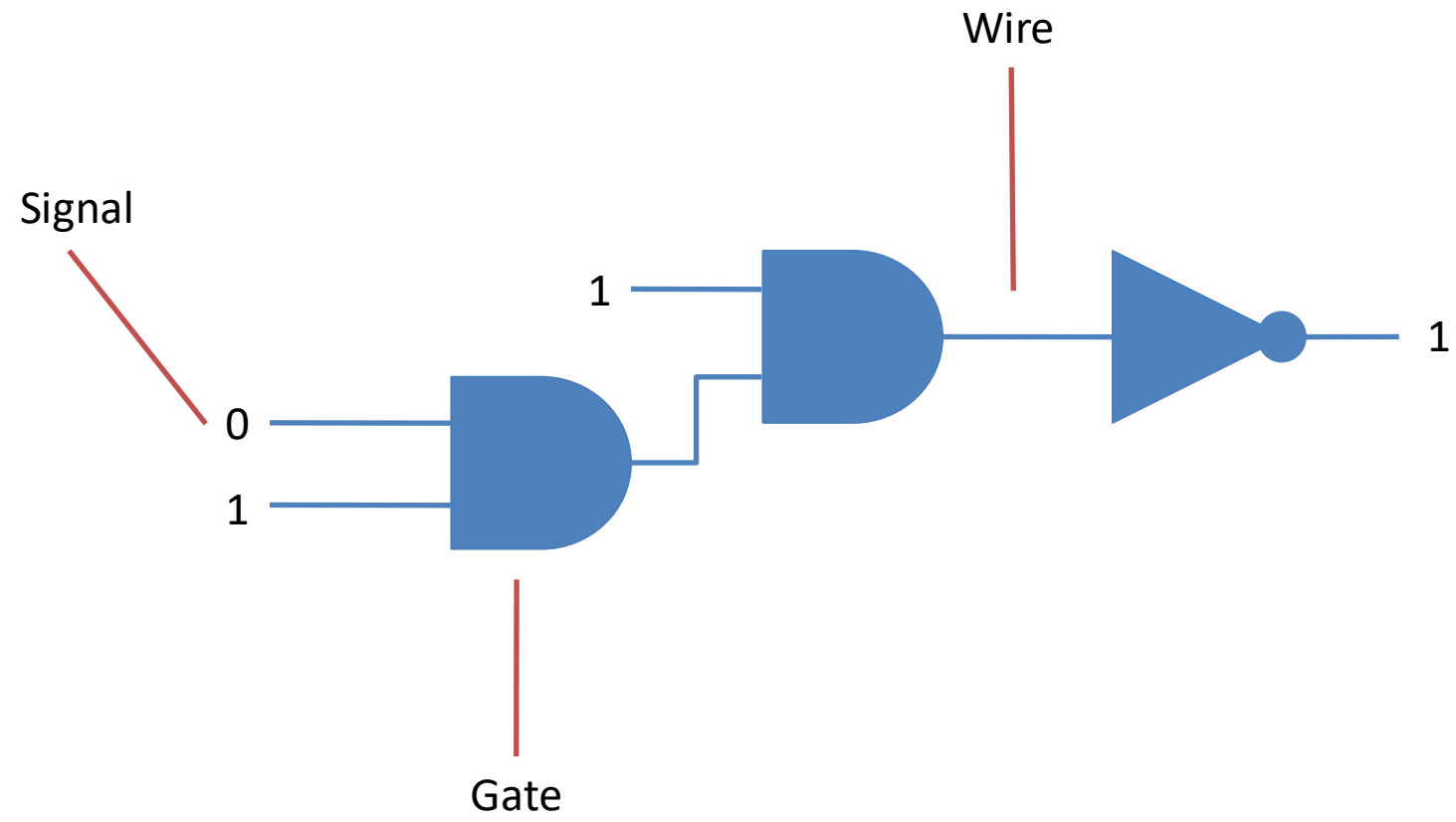
- We have seen how to build programs from assembly
- Now we'll learn how we implement assembly language instructions using circuits

A digital circuit is comprised of signals, gates, and wires

- Signals
 - Voltages applied to wires which generate electric current
- Binary signals are represented by different voltages:
 - 0: 0-1 volts
 - 1: 2-5 volts

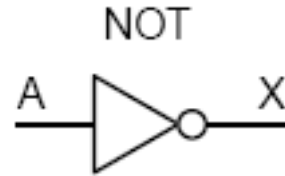
A digital circuit is comprised of signals, gates, and wires

- Gates
 - Devices which perform operations on signals corresponding to basic logic operations: and, or, not, nand, nor, xor
 - Made out of transistors
- Wires
 - Lines over which signals are transmitted between gates



Representation of Logic Gates

- Symbol

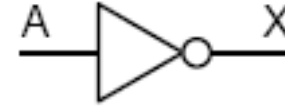


- Truth Table

A	X
0	1
1	0

- Algebraic Representation: $\overline{A}$
- Connection to mathematical logic: 0 = False, 1 = True

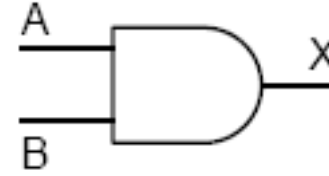
Not



A	X
0	1
1	0

- Inverts the input
- Algebraic representation: $\overline{A}$

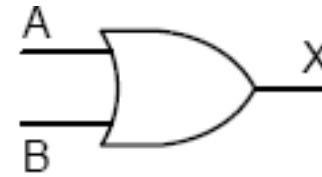
And



A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

- Algebraic representation: AB or $A \cdot B$
- Notice that multiplication makes sense here:
From the truth table, $X = A \cdot B$ with the usual meaning

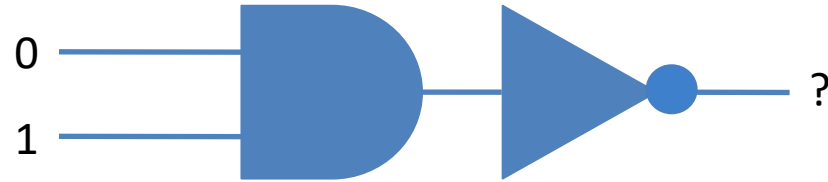
Or



A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

- Algebraic representation: $A+B$
- Addition is max as in $A + B = \max(A, B)$
- We'll see later that $+$ and $*$ obey some familiar properties of arithmetic (and some unfamiliar ones!)

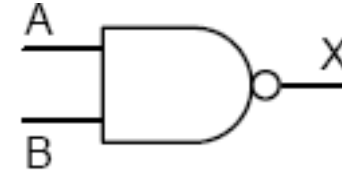
And and Not



A. 0

B. 1

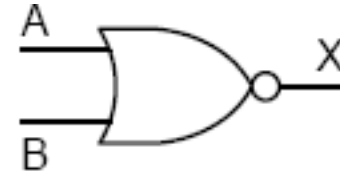
Nand



A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

- Algebraic representation: $\overline{AB}$
- Be careful with notation: $\overline{A \cdot B} = \overline{AB} \neq \overline{A} \cdot \overline{B}$

Nor



A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

- Algebraic representation: $\overline{A + B}$

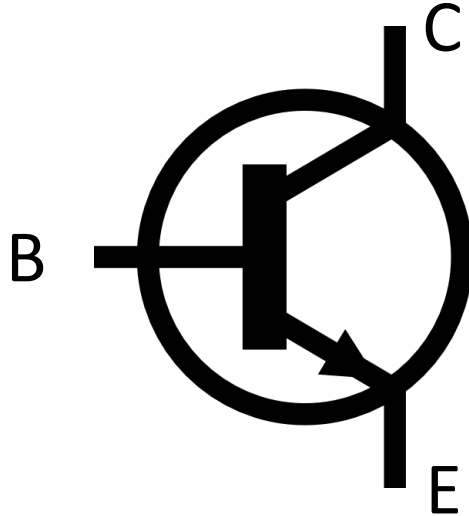
Xor



A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

- Algebraic representation: $A \oplus B$
- Programming languages tend to use $A \wedge B$
- Fun fact: XOR is addition without a carry

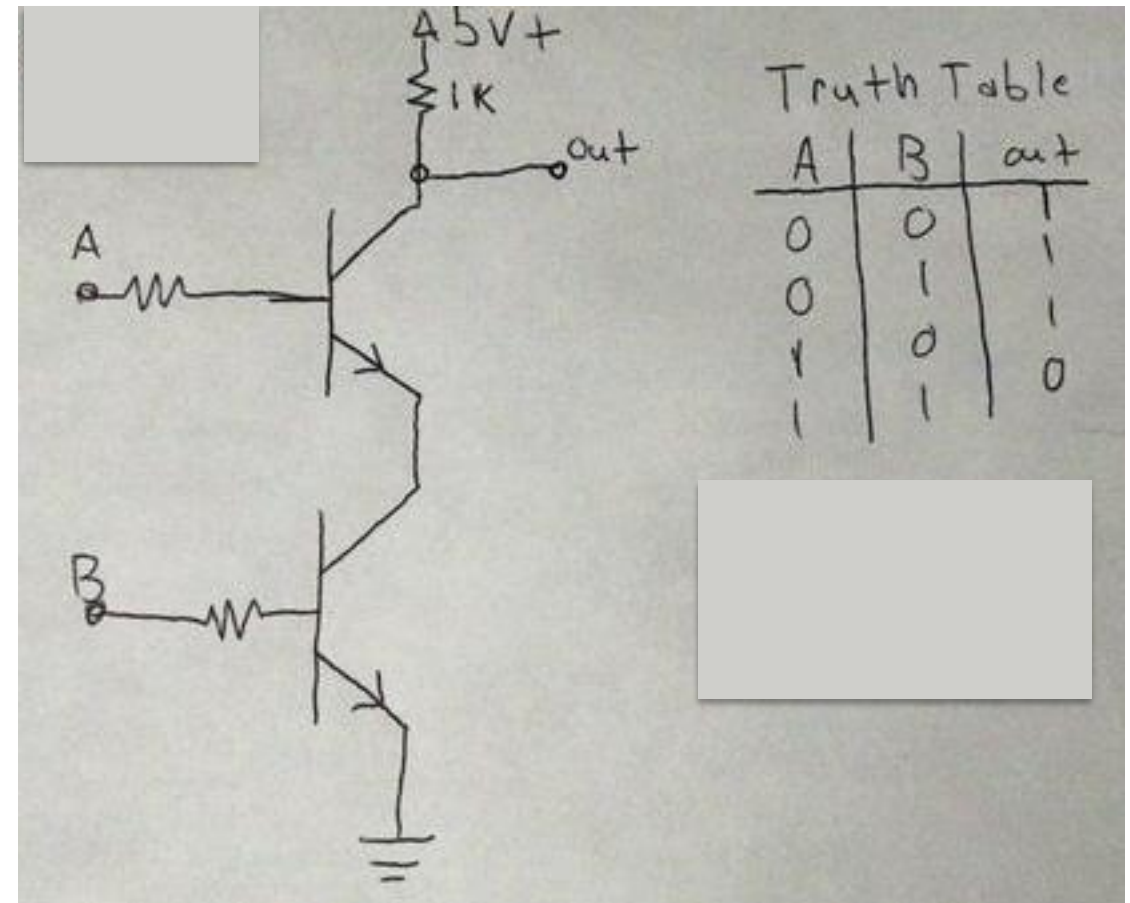
Our Friend the Transistor



- The basic electronic component from which all gates are created; there are many types, this is an NPN bipolar junction transistor (NPN BJT)
- Applying a voltage to the base (B) (relative to the emitter (E)) allows current to flow from the collector (C) to the emitter
- This creates an on/off switch

Building gates out of switches

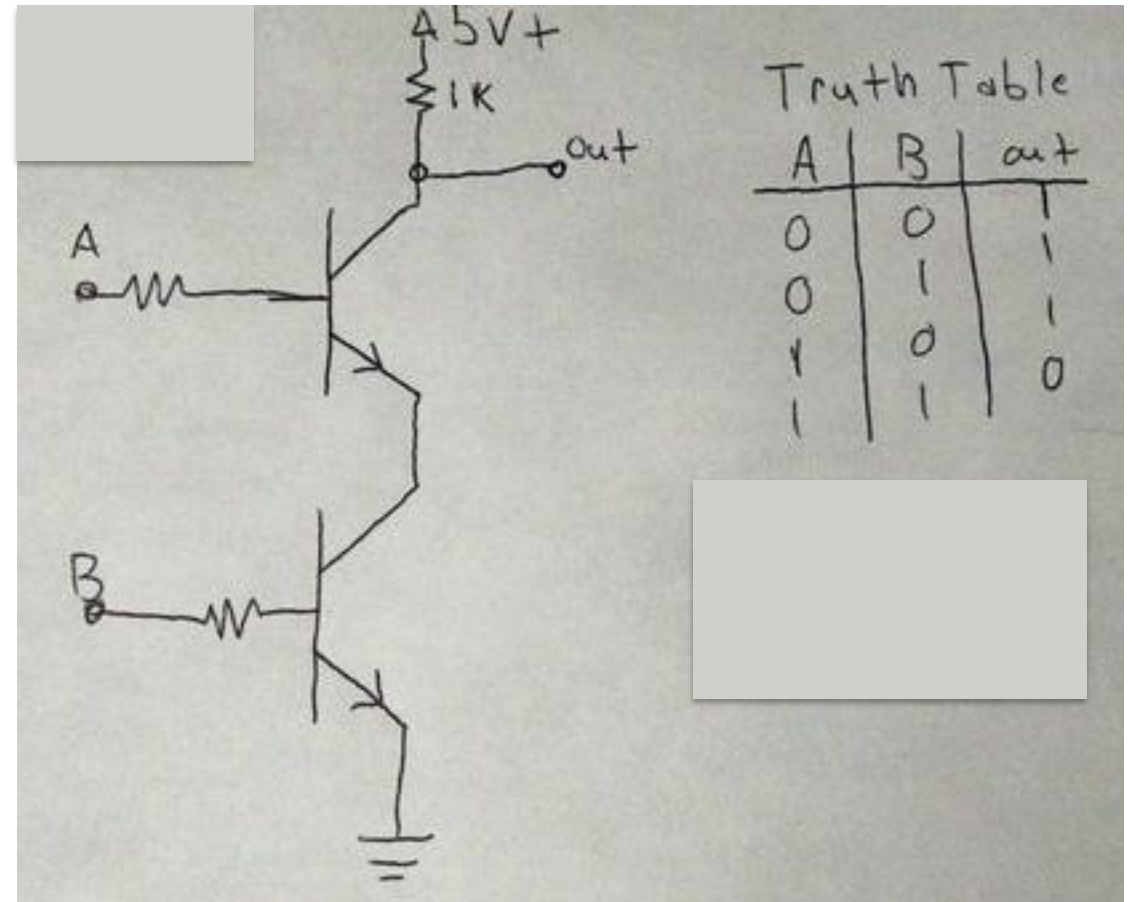
- Two inputs labeled A and B
- One output labeled out
- When A or B is 1, the other two electrodes (collector and emitter) are connected
- When A and B are both 1, out is connected to ground (logic value 0)
- When either A or B is 0, out is not connected to ground and current can flow from 5V to out

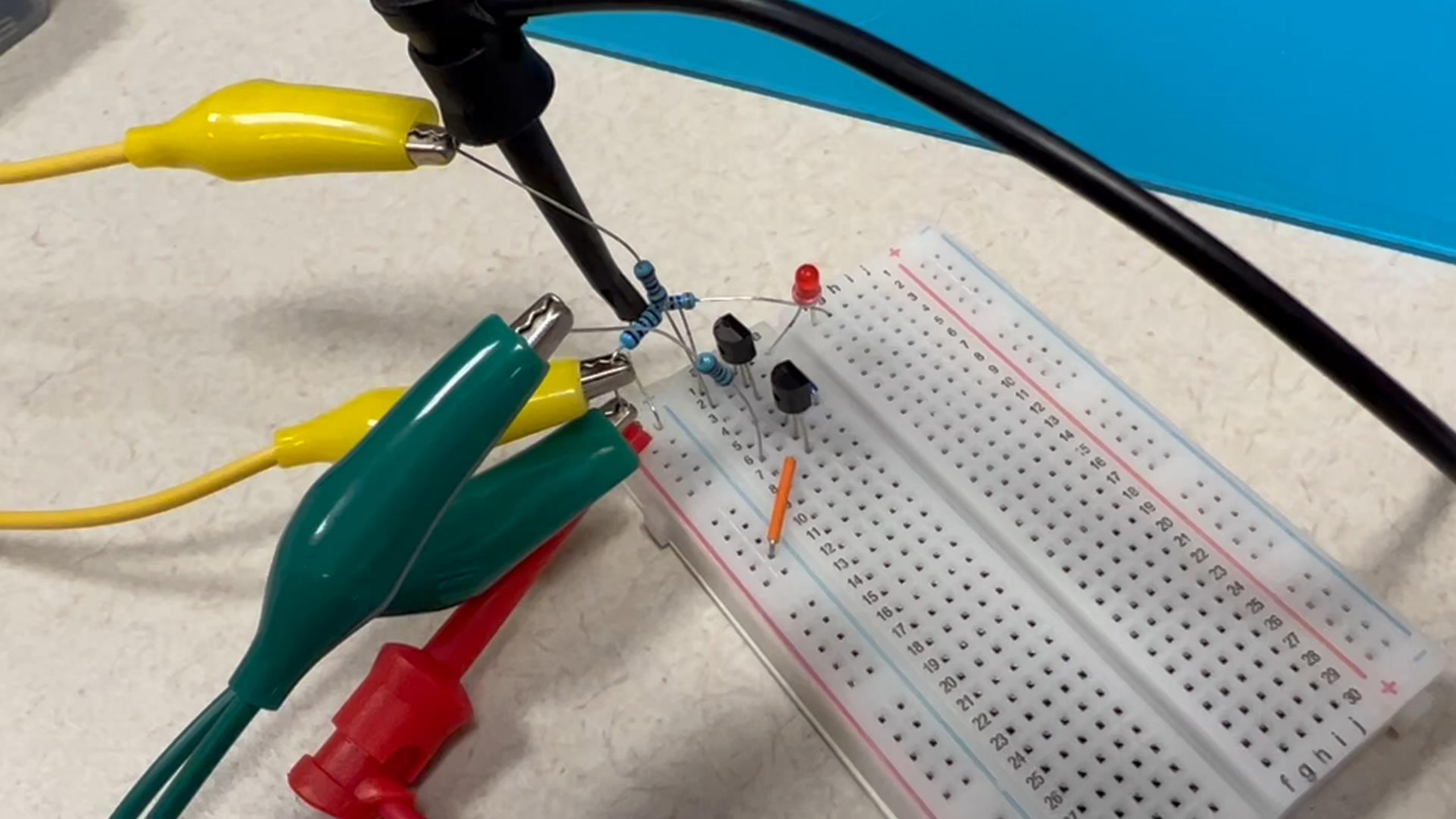


What Gate Does This Match?

- If both A and B are high voltage (logical 1), out will be low voltage (logical 0)
- Otherwise, out is high voltage

- A. AND
- B. OR
- C. NAND
- D. NOR





All Other Gates Can Be Created From NAND

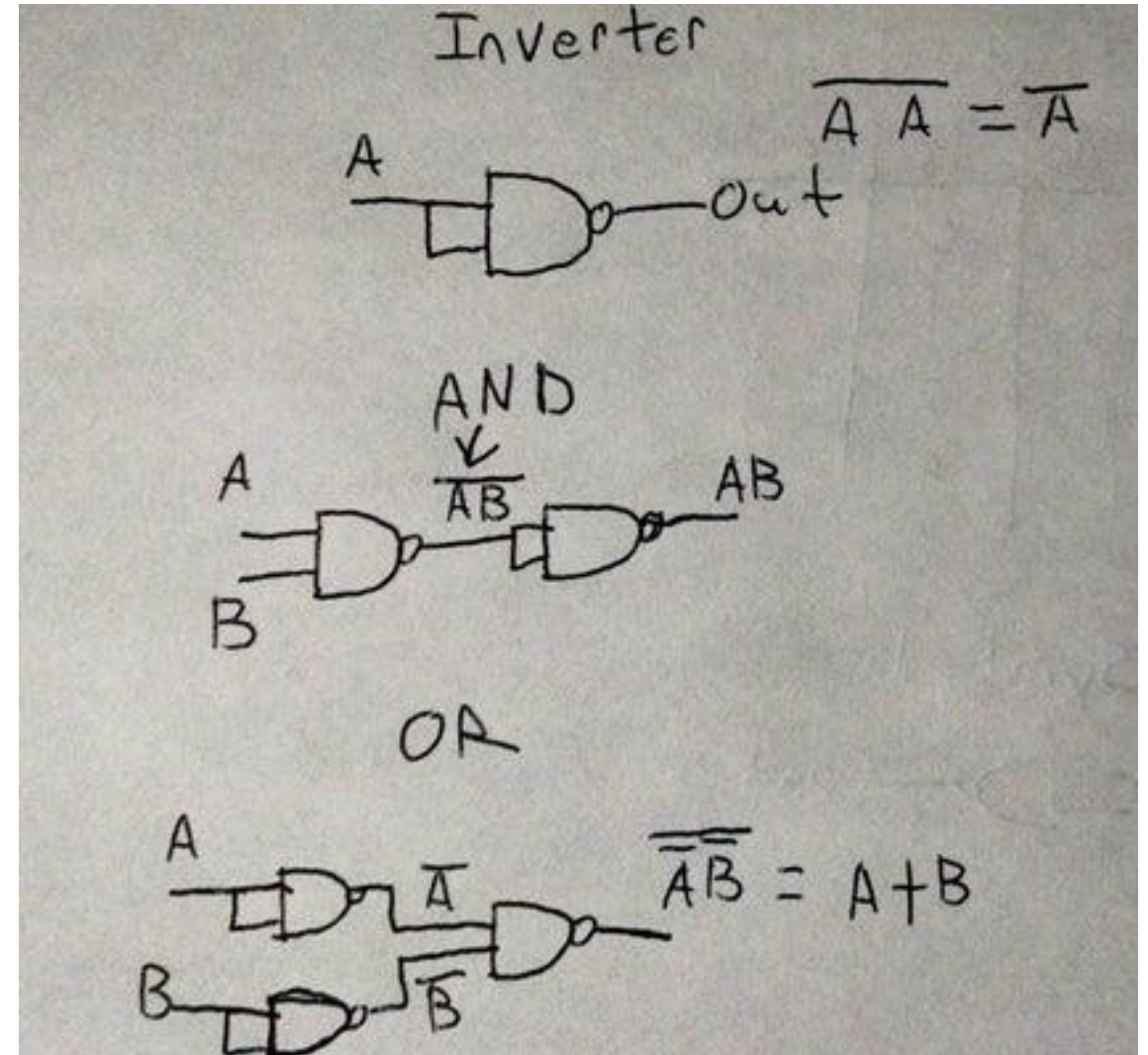
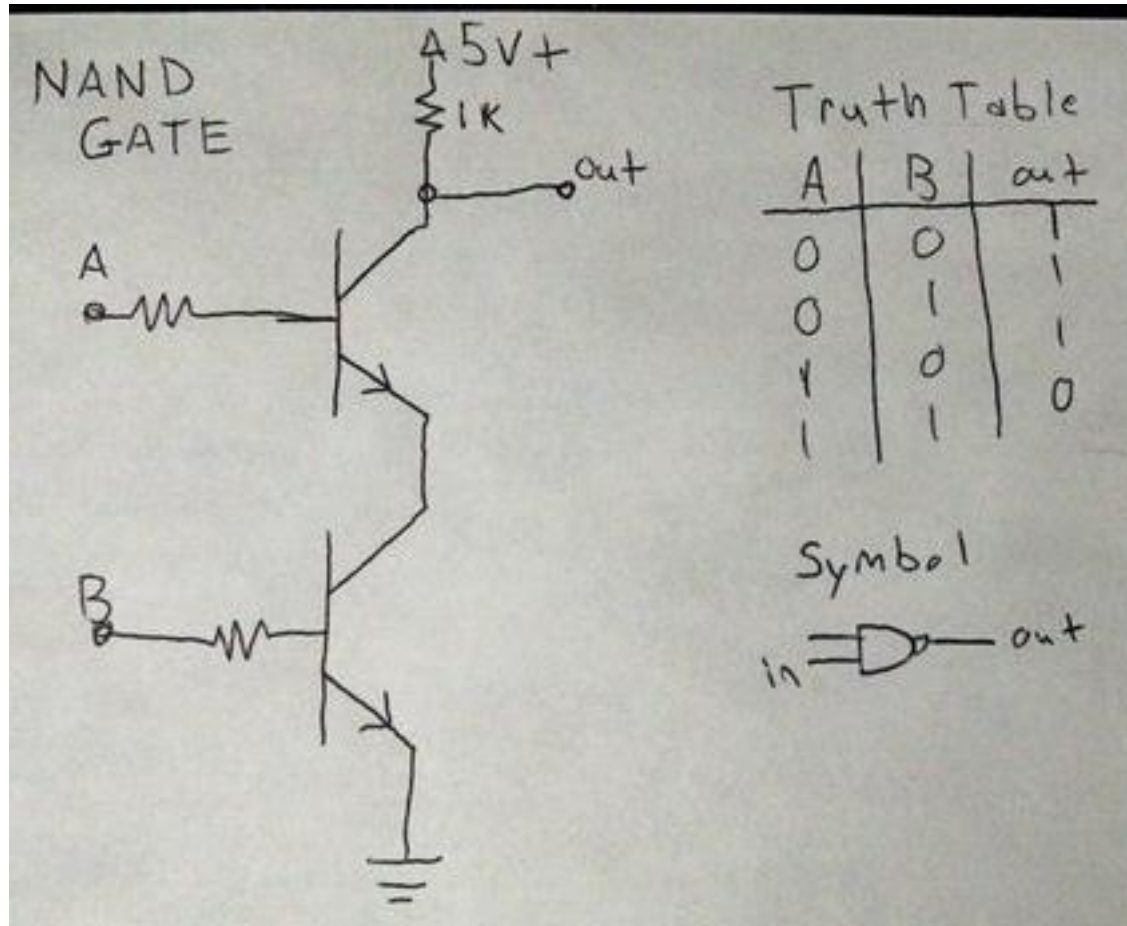
Not

And

Which is equivalent to $A \text{ OR } B$?

- A. $A \text{ NAND } B$
- B. $\text{NOT } (A \text{ NAND } B)$
- C. $(\text{NOT } A) \text{ NAND } (\text{NOT } B)$
- D. $\text{NOT } ((\text{NOT } A) \text{ NAND } (\text{NOT } B))$
- E. None of the above

Putting them together



All Gates Can Also Be Created from NOR

- NOR and NAND are universal gates
 - All gates can be created from them
- You will show this in Problem Set 5

Which column completes the truth table for

$$F = \overline{X} \cdot (Y + Z) ?$$

X	Y	Z	A	B	C	D
0	0	0	0	0	1	1
0	0	1	1	1	1	1
0	1	0	1	1	1	1
0	1	1	1	1	1	1
1	0	0	0	0	0	0
1	0	1	0	1	0	1
1	1	0	0	1	0	1
1	1	1	0	1	0	1

Groups: Draw circuit diagram for

$$F = \overline{X} \cdot (Y + Z)$$

$$F = \overline{A} + (B(AC + \overline{AB}))$$

Truth Table

A	B	C	AC	$\overline{AB}$	$AC + \overline{AB}$	$B(AC + \overline{AB})$	F
0	0	0					
0	0	1					
0	1	0					
0	1	1					
1	0	0					
1	0	1					
1	1	0					
1	1	1					